

Implementation of Inverse Hysteresis Model Into EMTP—Part II: Dynamic Model

Sergey E. Zirka, Yuriy I. Moroz, Nicola Chiesa, Robert G. Harrison, *Life Senior Member, IEEE*, and Hans Kr. Høidalen, *Senior Member, IEEE*

Abstract—The main features of a dynamic hysteresis model (DHM) and its implementation into the electromagnetic transient program (EMTP-ATP) are described. A method of fitting the DHM to catalog data is proposed. The implementation of the DHM within a transformer model is illustrated by transient calculations. Possible methods of fitting the DHM-based transformer model to a no-load test are outlined.

Index Terms—Electromagnetic Transients Program (EMTP), magnetic hysteresis, transformer cores.

I. INTRODUCTION

IT IS well known that the total loss in conducting ferromagnetic laminations depends not only on the flux density, but also on the flux waveform and frequency. Historically, the first method of modeling this dependence was to use the well-known Maxwell (penetration) equation for a thin ferromagnetic sheet [1] or its circuital (Foster, Cauer) equivalents [1], [2]. Providing a reliable basis for modeling nonoriented (NO) steels, as used mainly in generators and motors, this approach systematically undervalues the total loss in grain-oriented (GO) steels, which are the main core material of transformers, reactors, and many other devices. A distinguishing feature of GO material is the large percentage (40% and above [3]) of the so-called *excess loss* [4], which is usually missed by the commonly used finite-element Maxwell equation solvers. The term *excess loss* reflects the fact that this loss component is *anomalous* from the standpoint of the Classical Loss theory.

In this situation, simple three-component dynamic hysteresis models [5], [6] can be used to represent the core or its individual branches (limbs and yokes). In these composite models, the total

surface field $H(t)$ of the sheet is decomposed into hysteresis field $H_h(t)$, classical eddy-current field $H_{ec}(t)$, and excess field $H_{exc}(t)$

$$H(t) = H_h(t) + H_{ec}(t) + H_{exc}(t). \quad (1)$$

By multiplying (1) by an induction increment dB , one obtains a relationship for energy increments which, integrated over a cycle, constitute the total energy loss W (in J/m^3) in terms of its hysteresis, classical, and excess components [4]

$$W = W_h + W_{ec} + W_{exc}. \quad (2)$$

It is commonly accepted that the hysteresis loss W_h is quasi-static and, thus, frequency independent. The other right-hand terms in (2) have different frequency dependencies $W_{ec} \sim f$ and $W_{exc} \sim f^{1/2}$ [4], and so are represented separately in (2).

A suitable form of (1) for reproducing these frequency dependencies is the *dynamic hysteresis model* (DHM) [7], [8]

$$H(t) = H_h(B(t)) + k_{ec} \frac{dB}{dt} + G(B) \left| \frac{dB}{dt} \right|^\alpha \delta. \quad (3)$$

In the EMTP version considered here, the field $H_h(B)$ is found from the reversible *static hysteresis model* (SHM) [9] implemented in ATP in [10]. The constant $k_{ec} = d^2/(12\rho)$ is determined by the lamination thickness d and the resistivity ρ of the core material, and $\delta = \text{sign}(dB/dt) = \text{sign}(v)$ is a directional parameter (± 1). The constant α , typically ~ 0.5 , and the function $G(B)$ are fitting parameters of the DHM.

Examples of successful usage of the DHM (3) with sinusoidal and nonsinusoidal voltages $v(t)$ can be found in [7] and [8]. Curve $W(f)$ in Fig. 1 illustrates the applicability of the DHM to 0.26-mm-thick GO steel. This is Steel 1 in [7], tested in an Epstein frame with sinusoidal induction $B(t)$. This is the regime specified in catalogs of electrical steels and alloys. The dotted curve $W_{meas}(f)$ in Fig. 1 shows the measured losses.

Fig. 1 can also be used to demonstrate three common methods of accounting for core (or branch) loss. Straight line 1 illustrates the simplest and most frequently used method, where the nonlinear inductance of the core is shunted with a loss resistor R_{loss} , adjusted to provide a total loss for a prescribed regime. In Fig. 1, this is specified by point T. If, in addition, the static hysteretic properties of the core are taken into account, the value of R_{loss} is properly recalculated and the loss dependence becomes straight line 2. Finally, if the *whole* core loss is reproduced by a *static* hysteretic inductor, as in [11]–[13], the loss dependence is depicted by horizontal line 3. This means that these models are frequency independent and, therefore, are insensitive to the harmonic content of the voltage. Consequently,

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S. E. Zirka and Y. I. Moroz are with the Department of Physics and Technology, Dnepropetrovsk National University, Dnepropetrovsk 49050, Ukraine (e-mail: zirka@email.dp.ua).

N. Chiesa is with Statoil, Trondheim NO-7005, Norway (e-mail: nchie@statoil.com).

R. G. Harrison is with the Department of Electronics, Carleton University, Ottawa, ON K1S 5B6 Canada (e-mail: rgh@doe.carleton.ca).

H. K. Høidalen is with the Department of Electric Power Engineering, NTNU, the Norwegian University of Science and Technology, Trondheim N-7491, Norway (e-mail: hans.hoidalen@elkraft.ntnu.no).

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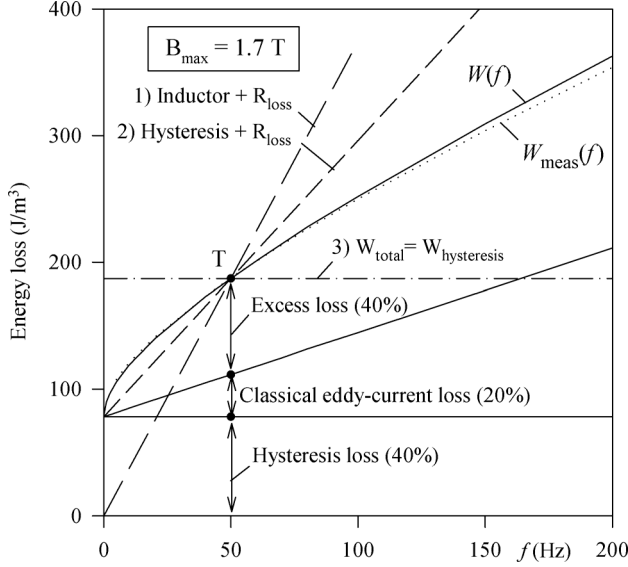


Fig. 1. Loss separation in GO steel with a sinusoidal induction ($B_{\max} = 1.7$ T).

as shown in [8], model (3) yields a substantial improvement over all other models mentioned.

In Part II of this paper, we describe an ATPDraw implementation of the DHM, which aims to reproduce the total loss of the steels considered and the shape of the corresponding dynamic loops.

II. IMPLEMENTATION OF THE DHM IN ATPDRAW

As in the case of the inverse SHM in [10], the *dynamic* hysteresis model is incorporated into ATPDraw as a hysteretic inductor using the Norton Type-94 MODELS block. This inductor is used to describe any individual magnetic branch of the magnetic circuit (for example, a transformer limb or yoke) of length l and cross section S . Referring the model parameters to a common number of turns N , the current in such a hysteretic inductor is, according to (1)

$$i(t) = \frac{l}{N} H(t) = i_h(t) + i_{ec}(t) + i_{exc}(t). \quad (4)$$

The dynamic hysteretic inductor corresponding to (3) and (4) is represented by the circuit in Fig. 2. The elements I_{hist} and g_h represent the static hysteretic inductor described in Part I. The linear conductance g_{ec} reproduces the current i_{ec} , while elements I_0 and g_{exc} characterize a nonlinear conductance carrying a current i_{exc} .

When solving the entire ATP network at the *current* time instant t , the parameters of the circuit at the *previous* time instant $t - \Delta t$ are used. Once voltage $v(t)$ is found, its value is used to recalculate all circuit parameters for time t as will be described.

The current $i_h(t)$ and the hysteretic conductance $g_h(t)$ are calculated using (10) and (11) in [10] where $H(t)$ is now understood to be $H_h(t)$.

From (3) and (4), the current in conductance g_{ec} is

$$i_{ec}(t) = \frac{l \cdot k_{ec}}{N} \frac{dB}{dt} = \frac{l \cdot k_{ec}}{N^2 S} v(t). \quad (5)$$

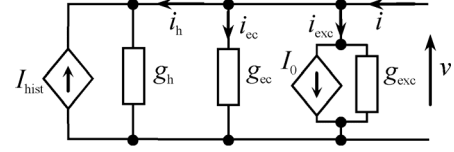


Fig. 2. Equivalent circuit of a dynamic hysteretic inductor.

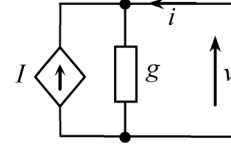


Fig. 3. Aggregate equivalent circuit of a dynamic hysteretic inductor.

A calculation of g_{ec} shows it to be a constant

$$g_{ec} = \frac{i_{ec}(t)}{v(t)} = \frac{l \cdot k_{ec}}{N^2 S} = \text{constant}. \quad (6)$$

In accordance with (3) and (4), the excess current is

$$i_{exc}(t) = \frac{l}{N} G(B) \left| \frac{dB}{dt} \right|^\alpha \delta = \frac{l \cdot G(B)}{N^{\alpha+1} S^\alpha} |v(t)|^\alpha \delta \quad (7)$$

with the derivation taking into account the equality $dB/dt = v(t)/(NS)$. The nonlinear dependence $i_{exc}(v)$ in (7) is approximated at $v(t)$ as $i_{exc}(v) = I_0 + g_{exc}v$, where

$$\begin{aligned} g_{exc}(t) &= \frac{di_{exc}(t)}{dv} = \frac{l}{N^{\alpha+1} S^\alpha} \frac{d}{dv} [G(B)|v(t)|^\alpha \delta] \\ &= \frac{l}{N^{\alpha+1} S^\alpha} \\ &\quad \times \left[\frac{dG(B)}{dB} \frac{d\lambda}{dv} \frac{1}{NS} |v(t)|^\alpha \delta + G(B)\alpha |v(t)|^{\alpha-1} \right]. \end{aligned} \quad (8)$$

When using the trapezoidal integration method, $d\lambda/dv = \Delta t/2$, the sought differential conductance is written as

$$\begin{aligned} g_{exc}(t) &= \frac{l}{N^{\alpha+1} S^\alpha} \\ &\quad \times \left[\frac{dG(B)}{dB} \frac{\Delta t}{2NS} |v(t)|^\alpha \delta + G(B)\alpha |v(t)|^{\alpha-1} \right]. \end{aligned} \quad (9)$$

Having found g_{exc} and i_{exc} , the current source I_0 in Fig. 2 is calculated as $I_0(t) = i_{exc}(t) - g_{exc}(t)v(t)$.

Since $\alpha < 1$, the second bracketed term in (9) and, thus, g_{exc} , becomes infinity when the voltage $v(t)$ crosses zero. Fortunately, there is no need to evaluate g_{exc} at small $v \sim dB/dt$ because, according to (7), the current i_{exc} approaches zero simultaneously with $|v(t)|$. So i_{exc} can be neglected if $|v(t)|$ falls below a threshold voltage $v_{\min}$. A value $v_{\min} = 10^{-2}$ V is employed in the model [14], where $g_{exc} = 0$ and $i_{exc} = 0$ if $|v(t)| < v_{\min}$.

The circuit in Fig. 2 is represented in ATP by the equivalent circuit of Fig. 3, where the total conductance $g(t)$ is the sum of all three conductances in Fig. 2; the resulting current source is $I(t) = I_{\text{hist}}(t) - I_0(t)$.

The Type-94 MODELS block described before is represented schematically by the same inductor group and icon as those in [10, Fig. 8]. The only difference is the name DHM used instead of the designation SHM in [10].

The list of controlled output variables, which are standard for a Type-94 block, are the current $i(t)$ at the end of the time step, the conductance $g(t)$, and the Norton source $I(t)$. It also includes the values of the fields H_h , $H_h + H_{ec}$, $H = H_h + H_{ec} + H_{exc}$, and the induction B in the inductor.

The user can monitor the evolution of $H(t)$ and $B(t)$ in an individual inductor during the magnetization process, and construct its static and dynamic hysteresis loops. These loops can be useful when evaluating the quality of the model as a whole, for example, in a transformer application.

For the user's convenience, the model in [14] contains parameters for four of the most-used steels. These include three GO steels: type M4 (27Z130 [15]); type M5 (35Z145 [15], [15]); type M4 as studied experimentally in [7]; and a typical 0.5-mm-thick NO steel, studied and modeled in [9]. According to loss data measured at 1.5 T and 50 Hz, this NO material is similar to EES-grade steel M400-50A.

In addition, the user can create an individual material data file. If the material reference number is set to zero in the window of the inductor block, then information about its static major loop, saturation curve, and model parameters can be read from the user-provided file named User_Steel.txt.

III. PRELIMINARY TESTING OF THE DHM IN ATPDRAW

The DHM implementation was first tested on the type-M4 GO steel studied in [7] (material 27Z130 in [15]). In [10], this steel was designated M4E.

In accordance with [7], steel M4E has a measured resistivity $\rho = 0.4824 \cdot 10^{-6} \Omega \cdot \text{m}$, lamination thickness $d = 0.26 \text{ mm}$, and parameter $\alpha = 0.5$. Function $G(B)$ from [16] was improved by using more degrees of freedom

$$G(B) = \begin{cases} G_m (1 + 0.103B_\delta^2 + 0.165B_\delta^4), & \text{for } B_\delta < 0, \\ G_m (1 + 0.191B_\delta^2 + 0.049B_\delta^4), & \text{for } B_\delta \geq 0. \end{cases} \quad (10)$$

Here, $G_m = 0.443$, $B_\delta = B \cdot \delta$, and $\delta = \text{sign}(v(t))$, which is ± 1 .

The ATP model was validated using a scheme with a single inductor. The inductor design parameters N , S , and l were all set to unity. In this case, the Weber-ampere curves of the inductor coincide with its $B(H)$ curves.

Steady-state dynamic hysteresis loops were calculated for sinusoidal voltages of frequencies 50 and 100 Hz, and four voltage (induction) amplitudes. Fig. 4 compares experimental and modeled loops as well as total losses W_{EXP} and W_{MOD} , as calculated from the corresponding loop areas.

IV. FITTING THE DHM TO CATALOG DATA FOR GO STEELS

Power losses (in watts per kilogram) for Nippon Steel Corporation steels 27Z130 and 35Z145, equivalent to steels M4 and M5, are given in [15], for 50 and 60 Hz and peak inductions B_m in the range 0.5 to 1.9 T.

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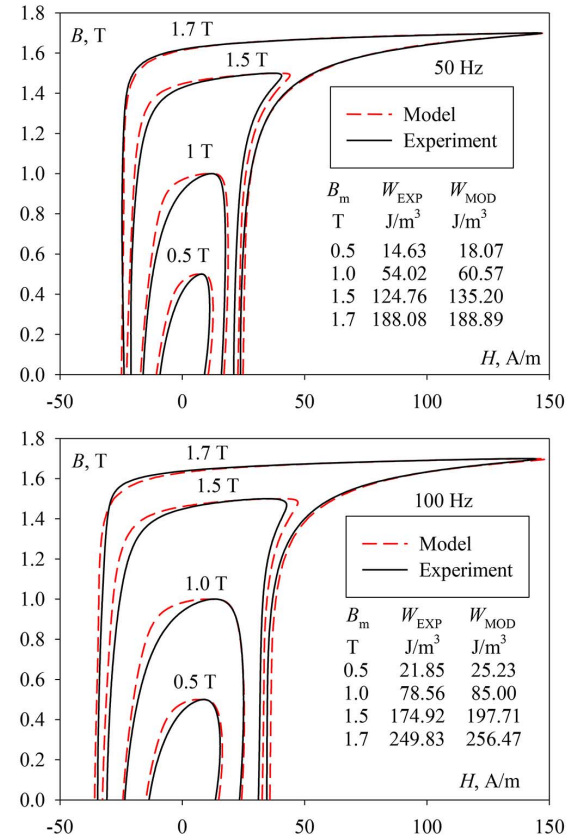


Fig. 4. Experimental and calculated dynamic loops of steel M4E (27Z130).

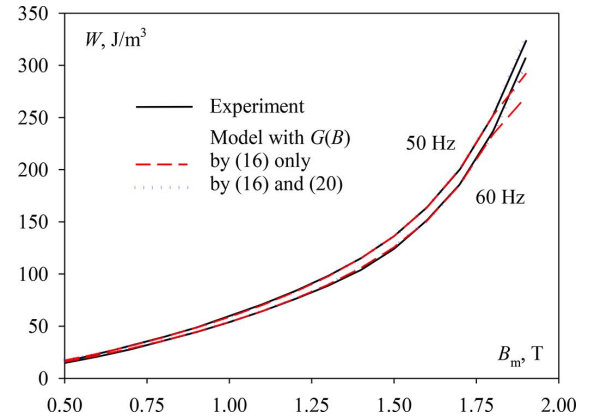


Fig. 5. Experimental (solid) and calculated (dashed) loss curves for steel M4. The inset shows the relative error ε incurred using (11) and (13). The dotted curves were obtained using (11), (13), and (15).

The solid curves in Fig. 5 use the same loss data for steel M4, but are recalculated in units of Joules per meter³. These units are more convenient for fitting the DHM since the area of a steady-state hysteresis loop in the $B - H$ plane is equal to the energy W (in Joules per meter³) dissipated per cycle and unit volume of the magnetic material.

Since the catalog [15] does not provide dynamic hysteresis loops (only the loss figures are given), we decided, following [8], to represent function $G(B)$ in the *symmetrical* form, in the

sense $G(-B) = G(B)$. This requirement is met by using polynomials in even powers of B only

$$G(B) = a_0 + a_1 B^2 + a_2 B^4 + \dots + a_n B^{2n}. \quad (11)$$

The value of n in (11) and the coefficients $a_i (i = 0, \dots, n)$ are fitting parameters.

The data in Fig. 5 were assembled into a table $W_{\text{exp}}(B_m, f)$ listing the energy losses for 15 consecutive values of peak induction B_m from 0.5 to 1.9 T, and at frequencies 50 and 60 Hz. Function (11) was fitted to this tabulated data as follows.

For every set of parameters $\mathbf{a} = [a_0, a_1, \dots, a_n]$, the double sum

$$F(\mathbf{a}) = \sum_{i=1}^2 \sum_{j=1}^{15} (W_{\text{exp}}(B_{mj}, f_i) - W_{\text{calc}}(B_{mj}, f_i, \mathbf{a}))^2 \quad (12)$$

was calculated for all flux densities and frequencies to obtain the cost function F . Here, $W_{\text{calc}}(B_{mj}, f_i, \mathbf{a})$ is the dynamic loop area calculated at the j th flux density and i th frequency.

For every chosen n in (11), minimization of function (12) was carried out on the parameters of vector $\mathbf{a}$ using the Nelder–Mead simplex method. The following “optimal” model parameters were found

$$\begin{aligned} n &= 3, & a_0 &= 0.352, & a_1 &= 0.08355, \\ a_2 &= -0.1945, & a_3 &= 0.1543. \end{aligned} \quad (13)$$

The dashed curves in Fig. 5 show calculated energy losses at 50 and 60 Hz, while the inset shows the relative error in the predicted loss, according to

$$\varepsilon(B_{mj}, f_i) = \frac{W_{\text{calc}}(B_{mj}, f_i, \mathbf{a}) - W_{\text{exp}}(B_{mj}, f_i)}{W_{\text{exp}}(B_{mj}, f_i)} 100\%. \quad (14)$$

The error in predicting magnetization losses in Fig. 5 does not exceed 3% either at $f = 50$ Hz for flux densities in the range 0.7 to 1.8 T, or at 60 Hz in the range 0.6 to 1.8 T. The increased error at $B_m = 1.9$ T is due to the fact that the major loop used in the model has a height of only 1.8 T, and that hysteresis phenomena at higher inductions were neglected, even though they do exist in reality. Taking into account that electrical devices, such as transformers, seldom operate at flux densities above 1.8 T, the modeling quality is considered acceptable. The missing static component of the hysteresis loss at $B_m > 1.8$ T can be compensated for by inclusion in function $G(B)$ of an additional term $(|B| - 1.8)a_{n+1}$. The modified function (11) then becomes

$$G(B) = G'(B) + a_{n+1}(|B| - 1.8), \text{ for } |B| > 1.8 \text{ T} \quad (15)$$

where $G'(B)$ is calculated using (11), or by using (16) for an asymmetrical $G(B)$. The dotted curves in Fig. 5 show calculated losses and the errors in their predicted values for Steel M4, when using (20) with $a_{n+1} = a_4 =$.

Fig. 6 shows dynamic loops calculated with (11) and (13) at 60 Hz. At $B > 1.5$ T, the loops in Fig. 6 are characterized by marked *widening* of their upper and lower (omitted) parts, which

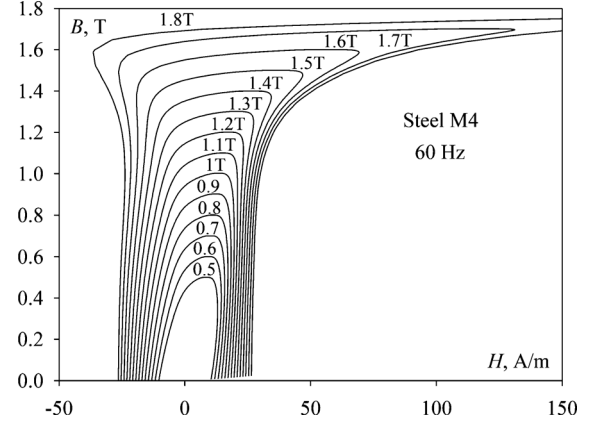


Fig. 6. Dynamic hysteresis loops for steel M4 calculated using a symmetrical fitting function $G(B)$ as determined by (11) and (13).

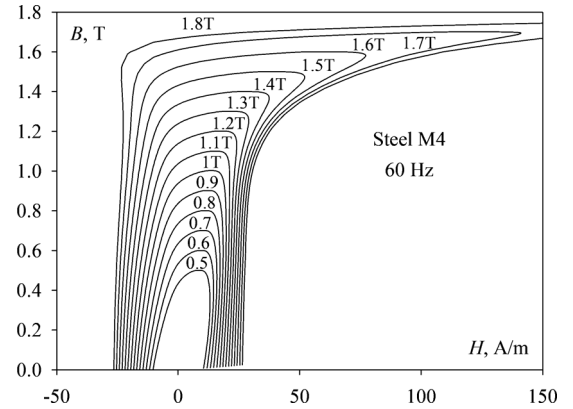


Fig. 7. Dynamic loops of steel M4 calculated with an asymmetrical function $G(B)$ defined by (16) and (13), using $q = 0.6$.

exceed the subtle experimentally observed widening shown in Fig. 4. To eliminate the excessive juts on the upper part of the descending branches (and the omitted lower part of the ascending branches), an asymmetry can be introduced into the fitting function $G(B)$

$$G(B) = \begin{cases} a_0 + (a_1 B^2 + \dots + a_n B^{2n})(1 - q) & \text{for } B_\delta < 0, \\ a_0 + (a_1 B^2 + \dots + a_n B^{2n})(1 + q) & \text{for } B_\delta \geq 0. \end{cases} \quad (16)$$

Here, the fitting parameter $q (< 1)$ determines the degree of asymmetry. For a given q , the decrease in $G(B)$ for $B_\delta < 0$ is the same as the increase in $G(B)$ for $B_\delta > 0$.

As a result, the dynamic loop area is kept unchanged. Fig. 7 shows dynamic hysteresis loops for steel M4 obtained using (16) with $q = 0.6$.

The dynamic loops in Fig. 7 seem more realistic than those in Fig. 6. Since the catalog [15] provides no dynamic hysteresis loops, we can make only a qualitative prediction of the loop shapes and so consider the model fitting for steel M4 to be complete.

In the case of steel M5 (35Z145), the coefficients in (11) found by minimizing cost function (12) are

$$\begin{aligned} n &= 3, & a_0 &= 0.4189, & a_1 &= -0.0269, \\ a_2 &= 0.191, & a_3 &= -0.0163. \end{aligned} \quad (17)$$

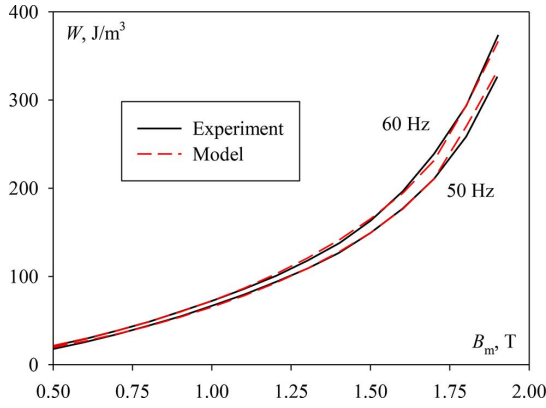


Fig. 8. Experimental (solid) and calculated (dashed) loss curves for steel M5. The inset shows the model error ε predicted using (15), (16) and (17) with $a_{n+1} = a_4 = 180$.

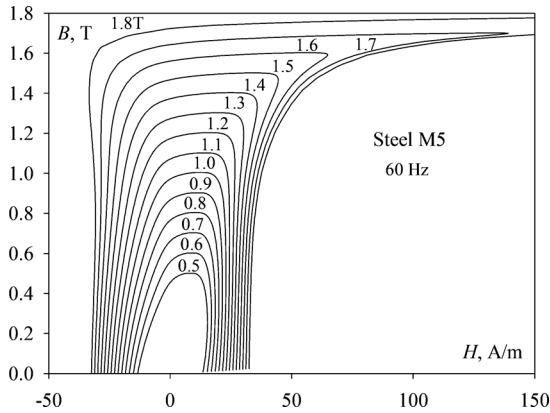


Fig. 9. Dynamic hysteresis loops for steel M5 calculated using the symmetrical function $G(B)$ defined by (11) and (17).

Fig. 8 shows catalog [15] (solid curves) and calculated (dashed curves) loss dependencies versus peak induction, for $f = 50$ and 60 Hz. For $|B| > 1.8$ T, expression (15) is used with $a_{n+1} = a_4 = 180$. The relative error does not exceed 5% in the range $0.6 < B < 1.9$ T, so the modeling as a whole is estimated to be successful.

Fig. 9 shows dynamic loops calculated with a *symmetrical* $G(B)$ function, using $q = 0$ in (16). This family of dynamic loops seems acceptable, so parameters (17) and $q = 0$ were chosen for the DHM fitting of Steel M5.

V. FITTING THE DHM TO EXPERIMENTAL DATA FOR NO STEEL

The NO steel considered has resistivity $\rho = 0.43 \times 10^{-6} \Omega \cdot \text{m}$ and sheet thickness $d = 0.5$ mm. Dynamic hysteresis loops were measured for controlled sinusoidal induction at $f = 50$ Hz and peak flux densities 0.5, 1.0, and 1.5 T. They are shown

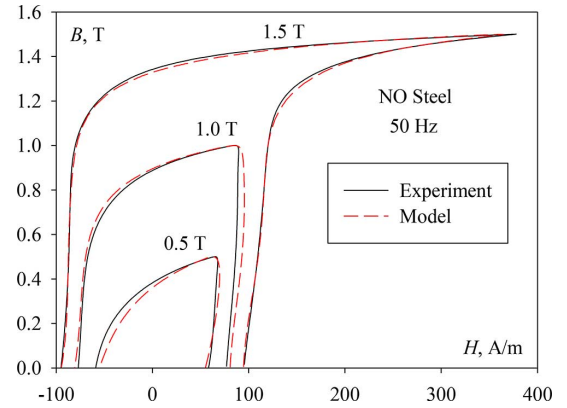


Fig. 10. Experimental and calculated dynamic loops of NO steel at 50 Hz, using the SHM with (20) and expressions (9) from [9].

in Fig. 10 as solid curves. The dynamic hysteresis model was fitted to the loop measured at amplitude of 1.5 T. Function $G(B)$ was represented as shown in (18) at the bottom of the page. By minimizing the deviation of the calculated dynamic loop from the experimental loop, the following set of the model parameters was found:

$$\begin{aligned} G_m &= 0.2526; a_{11} = 6.63225; a_{12} = 19.4563; \\ a_{13} &= 21.1076; a_{14} = 8.06186; a_{21} = 0.492104; \\ a_{22} &= 19.0235; a_{23} = -25.3507; a_{24} = 8.36722. \end{aligned} \quad (19)$$

As was found in the modeling, the dynamic loop calculated for $B_m = 1.5$ T almost coincides with the corresponding measured loop. However, as B_m is decreased, the area of the predicted loop becomes somewhat larger than that of the measured loop. This is due to the fact that the SHM of this steel was fitted to quasistatic FORCs, but not to static symmetrical loops, since we have no such loops for NO steel at our disposal.

In this situation, we used experimental *dynamic* loops in order to adjust the parameters of a *static* hysteresis model. As shown in Fig. 10, dynamic loops calculated at $B_m = 0.5$ and 1.0 T can be brought closer to experimental loops by applying a multiplier $\Delta B_{\text{out}}/(2B_T)$ to the polynomial of [9, (8)]. Here, ΔB_{out} is the height of the current minor loop and $2B_T$ is the major loop height.

Thus, parameter a for the SHM of the NO steel being modeled is calculated from the expression

$$a = \frac{\Delta B_{\text{out}}}{2B_T} \Delta B_{\text{rev}} (7.73 + 2.76\beta - 28.63\beta^2 + 28.36\beta^3). \quad (20)$$

As before, parameters b and c are calculated from [9, (9)]. The altered formula for parameter a does not affect the shape of calculated static FORCs.

$$G(B) = \begin{cases} G_m (1 + a_{11}B_\delta + a_{12}B_\delta^2 + a_{13}B_\delta^3 + a_{14}B_\delta^4) & \text{for } B_\delta < 0, \\ G_m (1 + a_{21}B_\delta + a_{22}B_\delta^2 + a_{23}B_\delta^3 + a_{24}B_\delta^4) & \text{for } B_\delta \geq 0. \end{cases} \quad (18)$$

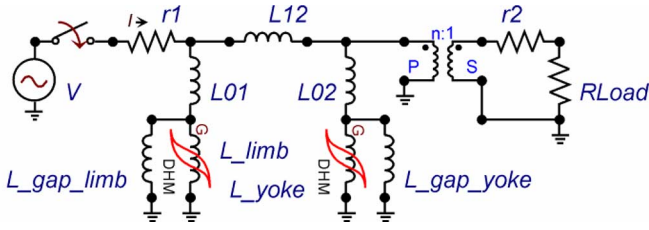


Fig. 11. Transformer model with linear inductances L_{gap_limb} and L_{gap_yoke} representing the core gaps.

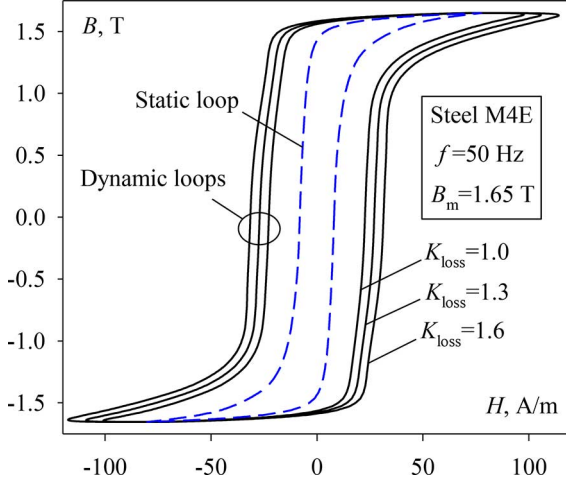


Fig. 12. Influence of K_{loss} on the shape of the dynamic loop.

VI. FITTING THE TRANSFORMER MODEL TO NO-LOAD LOSSES AND CURRENTS FROM OPEN-CIRCUIT TEST DATA

It is well known that transformer no-load losses and magnetizing currents normally exceed those measured for core material in an Epstein frame or a single-sheet tester.

This section shows that for transformer models based on the DHM (3), the no-load loss and current can be fitted *independently*. This is illustrated by using a single-phase transformer model [16] and, specifically, its duality-derived equivalent circuit in Fig. 11.

To adjust the model to the measured no-load loss, we multiply the constant k_{ec} and the function $G(B)$ in (3) by a loss coefficient K_{loss} , thereby altering the loss components W_{ec} and W_{exc} in (2).

The model is fitted to the open-circuit-test no-load current by introducing air gaps in the core. Individual gaps for the limbs and yokes can be included. In the equivalent circuit of Fig. 11, the limb gap is represented by linear inductance L_{gap_limb} in parallel with dynamic hysteretic inductor L_{limb} . The linear inductance L_{gap_yoke} in parallel with dynamic hysteretic L_{yoke} represents the yoke gap. Inductances L_{gap_limb} and L_{gap_yoke} are introduced in the transformer model externally.

For a gap thickness Δ , the equivalent inductance is calculated as

$$L_{gap} = \frac{N^2 S \mu_0}{\Delta} \quad (21)$$

where S is the cross-sectional area of the limb or yoke.

For simplicity, the limb and yoke gaps were set equal in the illustrative calculations below. The influences of K_{loss} and Δ are seen in Tables I and II, as well as in Figs. 12 and 13.

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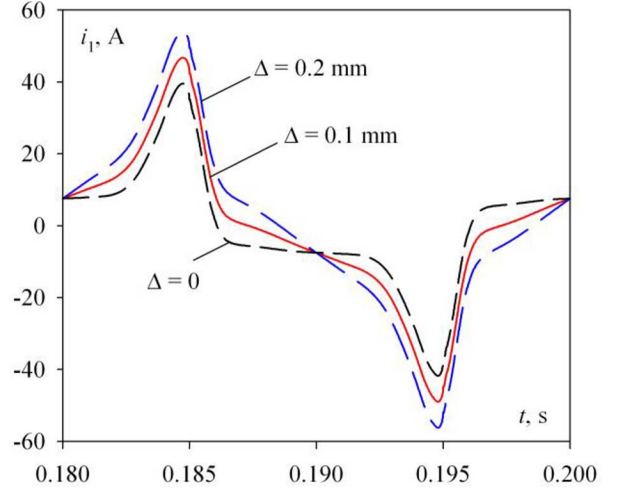


Fig. 13. Influence of air gap Δ on the waveshape of the transformer no-load current, assuming $K_{loss} = 1$.

TABLE I
TOTAL CORE LOSS (JOULES/CYCLE) VERSUS K_{loss} AND Δ

Δ , mm K_{loss}	0	0.05	0.1	0.15	0.2
1	2242.1	2239.2	2236.5	2234.1	2232.2
1.1	2377.3	2374.2	2371.5	2368.9	2366.7
1.2	2512.5	2509.2	2506.4	2503.7	2501.4
1.3	2647.6	2644.2	2641.3	2638.5	2636.1
1.4	2782.7	2779.3	2776.2	2773.3	2770.6
1.5	2917.8	2914.2	2911.1	2908.0	2905.2

TABLE II
RMS CURRENT (IN AMPERES) IN THE LV WINDING

Δ , mm K_{loss}	0	0.05	0.1	0.15	0.2
1	17.81	19.40	21.20	23.16	25.24
1.1	18.09	19.67	21.45	23.40	25.47
1.2	18.37	19.93	21.71	23.64	25.70
1.3	18.65	20.21	21.97	23.89	25.94
1.4	18.94	20.48	22.23	24.14	26.18
1.5	19.23	20.76	22.49	24.39	26.42

Tables I and II show that increasing K_{loss} substantially increases the total loss, but the rise of input current is moderate. On the other hand, increasing Δ leads to a substantial current rise, but almost unchanged losses. Thus, the losses and currents can be changed independently.

Fig. 14 demonstrates the separate evaluation of limb and yoke loops. The relevant dynamic loops in Fig. 14 represent dependencies $\lambda(i)$ where i are the currents in inductors L_{limb} and L_{yoke} (not in the linear inductances $L01$ and $L02$, which represent the flux paths in air [16]). For this reason, the loops Limb and Yoke are steeper than those calculated for the transformer as a whole (the latter loop takes into account the air gaps in the core).

The areas of the loops in Fig. 14 are total losses (in Joules) dissipated per cycle in the limb, yoke, and the entire transformer core.

Fig. 15 shows dynamic hysteresis loop during an inrush current event in the transformer model of Fig. 11, calculated using both hysteretic and saturation parts of the model simultaneously.

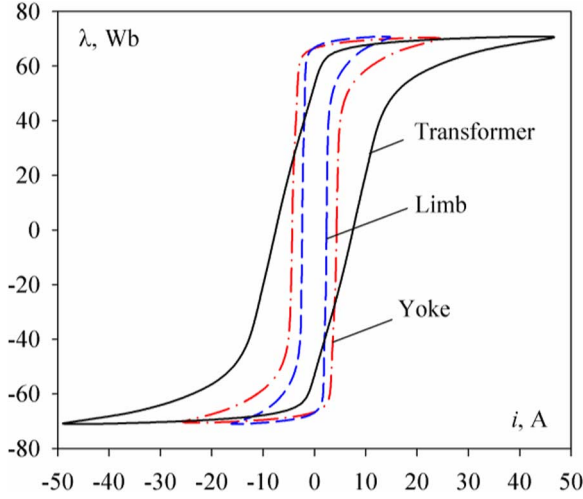


Fig. 14. Dynamic hysteresis loops for the limb, yoke, and the entire transformer, calculated using the transformer model of Fig. 11. The calculations assume $\Delta = 0.1$ mm.

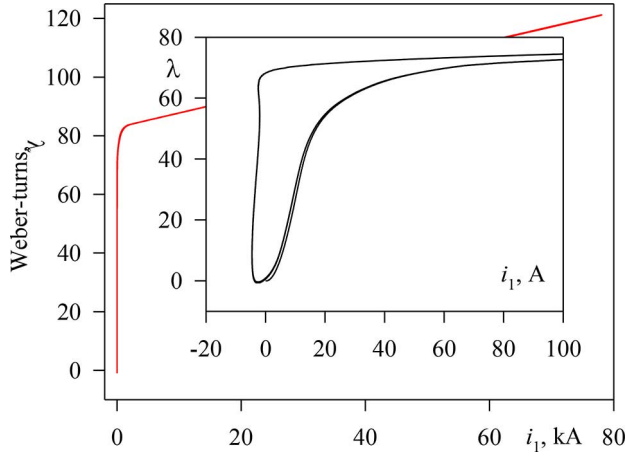


Fig. 15. Dynamic hysteresis loop during the inrush current event in the single-phase transformer model of Fig. 11.

The advantage of the DHM (3) over the existing DHMs [5], [6] or the simplest core model, which consists of a parallel-connected static hysteretic inductor and a constant loss resistor, is determined by the form of (3). In contrast to other DHMs, the final (excess-field) term in (3) is controlled by the function $G(B)$ which allows one to reproduce the dependence of the no-load transformer loss P_0 on the terminal voltage V . For example, curve 1 in Fig. 16 is a dependence $P_0(V)$ calculated for the transformer in [16] using

$$G(B) = G_m(1 + a_1 B^2 + a_2 B^4) \quad (22)$$

where $G_m = 0.4$, $a_1 = 0.4$, and $a_2 = 0.06$.

An increase of a_2 in (22) to 0.4 results in the loss curve 2 in Fig. 16. So by increasing the bracketed coefficients, or by including additional terms in $G(B)$, the loss curve can be made steeper, thus eliminating imperfections in the loss curves predicted by existing Electromagnetic Transients Program (EMTP) models (see [17, Fig. 4(b)]). By altering K_{loss} or G_m in (22), the loss curve can be shifted vertically, as shown by curve 3 in Fig. 16, calculated for $G_m = 0.5$.

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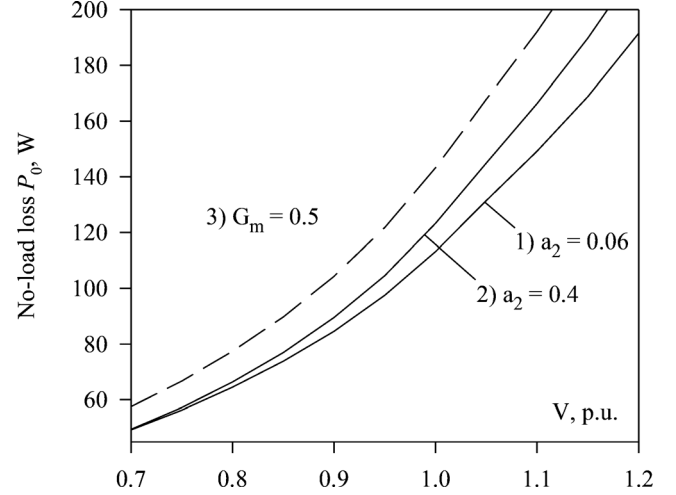


Fig. 16. No-load loss of the single-phase transformer model of Fig. 11.

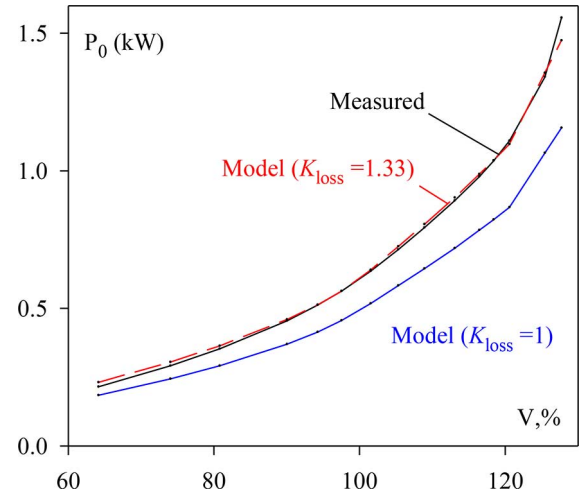


Fig. 17. No-load losses versus voltage calculated for a 300-kVA transformer.

If the core material and function $G(B)$ are chosen, the only fitting parameter of the DHM is the coefficient K_{loss} , while the core dimensions, including the air gaps Δ , are related to the transformer model.

Fig. 17 demonstrates the use of the DHM in the topological transformer model [18] where the DHM inductors are employed to represent the core legs and yokes individually. The no-load losses of a 300-kVA three-phase transformer calculated for $K_{\text{loss}} = 1$ and $K_{\text{loss}} = 1.33$ illustrate the influence of the loss coefficient.

Other details of the transformer modeling will be published elsewhere.

VII. CONCLUSION

This paper provides details of an implementation of a dynamic hysteresis model [8] in EMTP-ATP.

Previous works [9] and [8] have developed the static and dynamic hysteresis models for magnetic quantities, while this two-part paper converts these models into electrical quantities suitable for circuit analysis and, in addition, adapts the models to the trapezoidal integration method used in EMTP. At the same

time, the dynamic hysteresis model provides the user with the magnetic variables that allows one to control the conformity of $\lambda(i)$ curves used in EMTP with catalog $B-H$ curves. The paper also analyzes the characteristics and behavior of the model when subjected to power system disturbances.

The structure of the DHM (3), in particular, its variable final term, allows one to reproduce the shape of the dynamic hysteresis loop of a given material, and make its total loss dependent on the magnetization frequency and flux density. We describe the method of fitting the model to experimental or catalog data, and demonstrate its efficiency for typical electrical steels. The model fittings are provided in the model block in [14] allowing the ATP user to choose a steel from the list or to create a desired hysteretic inductor.

All of the tests performed confirm that the hysteretic inductor developed works correctly and that multiple instances of the model can be used without mutual interference.

This work demonstrates that the dynamic hysteresis model is suitable for incorporation into ATP electrical networks. An illustrative example can be found in [19], where the DHM is used in an ATPDraw model of a three-phase transformer.

Since the no-load loss and magnetization current of a given transformer differ from those calculated using the steel catalog data, a method is proposed to fit the transformer model to test data. For a chosen steel, the no-load transformer loss at a given terminal voltage is controlled by the single coefficient K_{loss} , whereas the no-load current is determined by the values of the air gaps in the core.

We also show the possibility of changing the shape of the dependence of the no-load loss curve on the terminal voltage. The details of this model-fitting procedure will be further elaborated and published elsewhere.

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Sergey E. Zirka received the Ph.D. and D.Sc. degrees in electrical engineering from the Institute of Electrodynamics, Kiev, Ukraine, in 1977 and 1992, respectively.

Since 1972, he has been with the Department of Physics and Technology, Dnepropetrovsk National University, Dnepropetrovsk, Ukraine, where he has been a Professor since 1992. His research interests include the modeling of magnetization processes in electrical steels, forming and transforming high-energy pulses, and transients in transformers of different types.

Yuriy I. Moroz received the Ph.D. degree in theoretical electrical engineering from the Institute of Modeling Problems in Energetics, the Ukrainian Academy of Sciences, Kiev, Ukraine, in 1991.

Currently, he is an Associate Professor in the Department of Physics and Technology, Dnepropetrovsk National University, Dnepropetrovsk, Ukraine. His research interests include the modeling of magnetization processes in electrical steels and transients in transformers of different types.

Nicola Chiesa received the M.Sc. degree in electrical engineering from Politecnico di Milano, Milano, Italy, in 2005 and the Ph.D. degree in electrical engineering from the Norwegian University of Science and Technology, Trondheim, Norway, in 2010.

From 2009 to 2013, he was a Research Scientist at SINTEF Energy Research, Trondheim, Norway, with a special interest in power transformers, transient simulations, power electronics, and energy-storage systems. In 2013, he joined Statoil, Trondheim, Norway, in the research-and-development department with a special interest in subsea power systems for oil and gas applications.

Robert G. Harrison (M'82–LM'08–LSM'09) received the B.A. and M.A. degrees in electrical engineering from Cambridge University, Cambridge, U.K., in 1956 and 1960, respectively, and the Ph.D. and D.I.C degrees in electrical engineering from the University of London, London, U.K., in 1964.

From 1964 to 1976, he was with the Research Laboratories of RCA Ltd., Ste-Anne-de-Bellevue, QC, Canada. In 1977, he became Director of Research at Com Dev Ltd., Dorval, QC. From 1979 to 1980, he was with Canadian Marconi Company, Montreal. Since 1980, he has been a Professor in the Department of Electronics, Carleton University, Ottawa, ON, Canada. He holds a number of basic patents on microwave frequency-division devices. His research interests include nonlinear microwave device/circuit interactions and physical models of ferromagnetic phenomena.

Prof. Harrison became a Distinguished Research Professor at Carleton University in 2005.

Hans Kr. Hoidalén (M'05–SM'14) was born in Norway in 1967. He received the M.Sc. and Ph.D. degrees in electrical engineering from the Norwegian University of Science and Technology, Trondheim, Norway, in 1990 and 1998, respectively.

Currently, he is a Professor at the Norwegian University of Science and Technology with a special interest in power system transients as well as electrical stress calculations and modeling. In recent years, the focus has been on transformer analysis and power system protection. He is the developer of the preprocessor to EMTP-ATP called ATPDraw.